1. 部分分式

將複雜的分式函數 $f(x)=\dfrac{P(x)}{Q(x)}$ 使用長除法與因式分解化簡成：$f(x)=S(x)+\dfrac{R(x)}{Q(x)}$，其中 $S(x)$ 為多項式 (當 $\deg P \geq \deg Q$ 才會有 $S(x)$ 這項)，而 $\frac{R(x)}{Q(x)}$ 為真分式 ($\deg R < \deg Q$). 對於真分式，依據代數基本定理，$Q(x)$ 必定可以拆為一次式與不可分解的二次式：

$$\frac{A_1}{x-a_1}+\cdots+\frac{A_m}{(x-a_m)^m}+\frac{B_1x+C_1}{x^2+b_1x+c_1}+\cdots+\frac{B_nx+C_n}{(x^2+b_nx+c_n)^n}$$

拆分後，每一項的積分通常簡單許多，從而簡化計算.

Question 使用部分分式拆解計算以下積分.

▷ $\displaystyle\int \frac{2x^4+3x^3+7x^2-5x-3}{x^3-x}\,\mathrm{d}x$

$$=\int 2x+3+\frac{9x^2-2x-3}{x^3-x}\,\mathrm{d}x=\int 2x+3+\frac{A}{x}+\frac{B}{x-1}+\frac{C}{x+1}\,\mathrm{d}x$$

所以：$9x^2-2x-3=A(x-1)(x+1)+Bx(x+1)+Cx(x-1)$

對照係數：$\begin{cases} A+B+C=9 \\ B-C=-2 \\ -A=-3 \end{cases} \Rightarrow (A,B,C)=(3,2,4)$

$$=\int\left(2x+3+\frac{3}{x}+\frac{2}{x-1}+\frac{4}{x+1}\right)\,\mathrm{d}x$$

$$=x^2+3x+3\ln|x|+2\ln|x-1|+4\ln|x+1|+C$$

▶ $\displaystyle\int \frac{3x^4+x^3+3x^2-3x-2}{x^3-x}\,\mathrm{d}x$

$$=\int 3x+1+\frac{6x^2-2x-2}{x^3-x}\,\mathrm{d}x=\int 3x+1+\frac{A}{x}+\frac{B}{x-1}+\frac{C}{x+1}\,\mathrm{d}x$$

所以：$6x^2-2x-2=A(x-1)(x+1)+Bx(x+1)+Cx(x-1)$

對照係數：$\begin{cases} A+B+C=6 \\ B-C=-2 \\ -A=-2 \end{cases} \Rightarrow (A,B,C)=(2,1,3)$

$$=\int\left(3x+1+\frac{2}{x}+\frac{1}{x-1}+\frac{3}{x+1}\right)\,\mathrm{d}x$$

$$=\frac{3}{2}x^2+x+2\ln|x|+\ln|x-1|+3\ln|x+1|+C$$

2. 變數變換

變數變換的本質，其實就是微分鏈鎖律（Chain Rule）在積分學中的逆向應用. 它除了能在代數上幫我們化簡複雜的積分計算外，更直觀地告訴我們：當我們換掉變數時，對應微分量（$\mathrm{d}x$ 與 $\mathrm{d}u$）之間的比例關係[1]：

Theorem. (**變數變換法（The Substitution Rule）**) 若 $u = g(x)$ 是一個值域為 I 的可微函數，且函數 f 在 I 上連續，那麼：

$$\int f(g(x))g'(x)\,\mathrm{d}x = \int f(u)\,\mathrm{d}u$$

Question 計算以下積分式.

▷ $\displaystyle\int 2x\sqrt{x^2+1}\,\mathrm{d}x = \int \sqrt{x^2+1}\,(2x\,\mathrm{d}x)$ 令 $u = x^2+1$，則 $\mathrm{d}u = 2x\mathrm{d}x$

$$= \int \sqrt{u}\,\mathrm{d}u$$
$$= \frac{u^{3/2}}{3/2} + C = \frac{2}{3}(x^2+1)^{3/2} + C$$

▷ $\displaystyle\int_0^1 \frac{x^4}{x^5+1}\,\mathrm{d}x = \int_0^1 \frac{1}{x^5+1}\,(x^4\,\mathrm{d}x)$ 令 $u = x^5+1$，則 $\mathrm{d}u = 5x^4\mathrm{d}x$

$$= \int_1^2 \frac{1}{5u}\,\mathrm{d}u$$
$$= \frac{1}{5}\ln|u|\Big|_1^2 = \frac{1}{5}\ln 2$$

▶ $\displaystyle\int x^3\cos(x^4+2)\,\mathrm{d}x = \int \cos(x^4+2)\,(x^3\,\mathrm{d}x)$ 令 $u = x^4+2$，則 $\mathrm{d}u = 4x^3\mathrm{d}x$

$$= \int \frac{1}{4}\cos u\,\mathrm{d}u$$
$$= \frac{1}{4}\sin u + C = \frac{1}{4}\sin(x^4+2) + C$$

▶ $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \tan x\,\mathrm{d}x = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x}{\cos x}\,\mathrm{d}x$ 令 $u = \cos x$，則 $\mathrm{d}u = -\sin x\mathrm{d}x$

$$= \int_{\frac{\sqrt{3}}{2}}^{\frac{1}{2}} -\frac{1}{u}\,\mathrm{d}u$$
$$= -\ln|u|\Big|_{\frac{\sqrt{3}}{2}}^{\frac{1}{2}} = -\ln\left(\frac{1}{2}\right) - \left(-\ln\left(\frac{\sqrt{3}}{2}\right)\right) = \ln\sqrt{3} = \frac{1}{2}\ln 3$$

[1]在單變數微積分中，這個縮放比例就是導數 $g'(x)$；推廣到多變數微積分時，則是由雅可比行列式（Jacobian Determinant）得到微分量之間的關係.

2.1. 三角代換

三角代換算式變數變換的一種特殊方法，三角代換是為了利用三角恆等式替換根號內的平方和或平為 $\sin\theta$、$\tan\theta$、$\sec\theta$：

積分中的表達式	代換方法	代換結果
$\sqrt{a^2-x^2}$	$x=a\sin\theta,\ -\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$	$1-\sin^2\theta=\cos^2\theta$
$\sqrt{a^2+x^2}$	$x=a\tan\theta,\ -\frac{\pi}{2}<\theta<\frac{\pi}{2}$	$1+\tan^2\theta=\sec^2\theta$
$\sqrt{x^2-a^2}$	$x=a\sec\theta,\ 0\le\theta<\frac{\pi}{2}$ or $\pi\le\theta<\frac{3\pi}{2}$	$\sec^2\theta-1=\tan^2\theta$

Question 計算以下積分式.

▷ $$\int\frac{1}{1+x^2}\,\mathrm{d}x=\int\frac{1}{1+x^2}\,\mathrm{d}x \qquad \text{令 } x=\tan u\text{，則 } \mathrm{d}x=\sec^2 u\mathrm{d}u$$
$$=\int\frac{1}{1+\tan^2 u}\cdot\sec^2 u\,\mathrm{d}u=\int 1\,\mathrm{d}u$$
$$=u+C=\tan^{-1}x+C$$

▷ $$\int\frac{1}{x^2+x+2}\,\mathrm{d}x=\int\frac{1}{\left(x+\frac{1}{2}\right)^2+\frac{7}{4}}\,\mathrm{d}x \qquad \text{令 } x+\frac{1}{2}=\frac{\sqrt{7}}{2}\tan u\text{，則 } \mathrm{d}x=\frac{\sqrt{7}}{2}\sec^2 u\mathrm{d}u$$
$$=\int\frac{1}{\frac{7}{4}\tan^2 u+\frac{7}{4}}\cdot\frac{\sqrt{7}}{2}\sec^2 u\,\mathrm{d}u=\int\frac{2}{\sqrt{7}}\,\mathrm{d}u$$
$$=\frac{2}{\sqrt{7}}u+C=\frac{2}{\sqrt{7}}\tan^{-1}\left(\frac{2x+1}{\sqrt{7}}\right)+C$$

▶ $$\int\frac{1}{\sqrt{4-x^2}}\,\mathrm{d}x=\int\frac{1}{\sqrt{4-x^2}}\,\mathrm{d}x \qquad \text{令 } x=2\sin u\text{，則 } \mathrm{d}x=2\cos u\mathrm{d}u$$
$$=\int\frac{1}{\sqrt{4-4\sin^2 u}}\cdot 2\cos u\,\mathrm{d}u=\int 1\,\mathrm{d}u$$
$$=u+C=\sin^{-1}\left(\frac{x}{2}\right)+C$$

Remark 只要記住「根號內的數值必須大於等於零」，或者分別嘗試 sin、tan、sec 看哪個可以成功湊出恆等式.

3. 分部積分

微分乘法法則告訴我們：

$$\frac{\mathrm{d}}{\mathrm{d}x}(F(x)\cdot G(x)) = F(x)g(x) + f(x)G(x)$$

其中 $F'(x) = f(x)$、$G'(x) = g(x)$. 同時對兩邊積分可以得到：

$$F(x)\cdot G(x) = \int F(x)g(x)\,\mathrm{d}x + \int f(x)G(x)\,\mathrm{d}x$$

移項後有：

Theorem. **(分部積分法（Integration by Parts, IBP）)**

$$\int F(x)g(x)\,\mathrm{d}x = F(x)\cdot G(x) - \int f(x)G(x)\,\mathrm{d}x$$

Question 使用 IBP 化簡以下積分式.

▷ $\displaystyle\int x\ln x\,\mathrm{d}x = x^2\frac{\ln x}{2} - \int \frac{x}{2}\,\mathrm{d}x$

▷ $\displaystyle\int \ln x\,\mathrm{d}x = x\ln x - \int 1\,\mathrm{d}x$

▶ $\displaystyle\int xe^x\,\mathrm{d}x = xe^x - \int e^x\,\mathrm{d}x$

▶ $\displaystyle\int \tan^{-1}(2x)\,\mathrm{d}x = x\tan^{-1}(2x) - \int x\cdot\frac{2}{1+4x^2}\,\mathrm{d}x$

▶ $$\begin{aligned}\int e^x\sin x\,\mathrm{d}x &= e^x\sin x - \int e^x\cos x\,\mathrm{d}x\\ &= e^x\sin x - \left(e^x\cos x - \int e^x(-\sin x)\,\mathrm{d}x\right)\\ &= e^x(\sin x - \cos x) - \int e^x\sin x\,\mathrm{d}x\\ \Rightarrow\ 2\int e^x\sin x\,\mathrm{d}x &= e^x(\sin x - \cos x)\\ \therefore \int e^x\sin x\,\mathrm{d}x &= \frac{e^x}{2}(\sin x - \cos x) + C\end{aligned}$$

Remark 可以發現，分部積分使用時機大致有：1. 多項式乘上指數函數、對數函數、三角函數、反三角函數，並且微分對象優先嘗試順序為對數、反三角、多項式、三角、指數. 2. 某函數微分有循環的，如 e^x、$\sin(x)$、$\cos(x)$，有機會可以回到原積分式.

4. 瑕積分

Definition 若積分上、下限中有一端為無窮，則稱該積分屬於**第一類瑕積分**：

- $\displaystyle\int_a^{\infty} f(x)\,\mathrm{d}x = \lim_{t\to\infty}\int_a^{t} f(x)\,\mathrm{d}x$
- $\displaystyle\int_{-\infty}^{b} f(x)\,\mathrm{d}x = \lim_{t\to-\infty}\int_t^{b} f(x)\,\mathrm{d}x$
- $\displaystyle\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \int_{-\infty}^{a} f(x)\,\mathrm{d}x + \int_a^{\infty} f(x)\,\mathrm{d}x$

Remark 對應的極限若存在，稱該積分收斂 (Convergent)；若不存在，則稱發散 (Divergent).

Question 請利用定義將下列瑕積分改寫為定積分的極限式.

▷ $\displaystyle\int_1^{\infty} 2x^{-3}\,\mathrm{d}x = \lim_{t\to\infty}\int_1^{t} 2x^{-3}\,\mathrm{d}x$

$$= \lim_{t\to\infty}\left[-x^{-2}\right]_1^{t}$$

$$= \lim_{t\to\infty}(-t^{-2}+1) = 1$$

▷ $\displaystyle\int_{-\infty}^{\infty}\frac{1}{x^2+1}\,\mathrm{d}x = \lim_{t\to-\infty}\int_t^{0}\frac{1}{x^2+1}\,\mathrm{d}x + \lim_{s\to\infty}\int_0^{s}\frac{1}{x^2+1}\,\mathrm{d}x$

$$= \lim_{t\to-\infty}\left[\tan^{-1}x\right]_t^{0} + \lim_{s\to\infty}\left[\tan^{-1}x\right]_0^{s}$$

$$= \left(0-\left(-\frac{\pi}{2}\right)\right) + \left(\frac{\pi}{2}-0\right) = \pi$$

▶ $\displaystyle\int_{-\infty}^{-1}\frac{1}{\sqrt[3]{x}}\,\mathrm{d}x = \lim_{t\to-\infty}\int_t^{-1} x^{-1/3}\,\mathrm{d}x$

$$= \lim_{t\to-\infty}\left[\frac{3}{2}x^{2/3}\right]_t^{-1}$$

$$= \frac{3}{2} - \lim_{t\to-\infty}\frac{3}{2}t^{2/3} = \infty \text{ (發散)}$$

▶ $\displaystyle\int_{-\infty}^{\infty}\frac{1}{x^2+9}\,\mathrm{d}x = \lim_{t\to-\infty}\int_t^{0}\frac{1}{x^2+9}\,\mathrm{d}x + \lim_{s\to\infty}\int_0^{s}\frac{1}{x^2+9}\,\mathrm{d}x$

$$= \lim_{t\to-\infty}\left[\frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right)\right]_t^{0} + \lim_{s\to\infty}\left[\frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right)\right]_0^{s}$$

$$= \frac{1}{3}\left(0-\left(-\frac{\pi}{2}\right)\right) + \frac{1}{3}\left(\frac{\pi}{2}-0\right) = \frac{\pi}{3}$$

Definition 若被積函數在積分區間內某點不連續或無界，則該積分屬**第二類瑕積分**：

- $\displaystyle\int_a^b f(x)\,\mathrm{d}x = \lim_{t\to b^-}\int_a^t f(x)\,\mathrm{d}x$ (當 $f(x)$ 在 $x=b$ 不連續)
- $\displaystyle\int_a^b f(x)\,\mathrm{d}x = \lim_{t\to a^+}\int_t^b f(x)\,\mathrm{d}x$ (當 $f(x)$ 在 $x=a$ 不連續)
- $\displaystyle\int_a^b f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x + \int_c^b f(x)\,\mathrm{d}x$ (當 $f(x)$ 在 (a,b) 內某點 c 不連續)

Question 請利用定義將下列瑕積分改寫為定積分的極限式.

▷ $$\begin{aligned}\int_0^1 \frac{1}{x}\,\mathrm{d}x &= \lim_{t\to 0^+}\int_t^1 \frac{1}{x}\,\mathrm{d}x\\ &= \lim_{t\to 0^+}\left[\ln x\right]_t^1\\ &= \lim_{t\to 0^+}(0-\ln t) = \infty \text{ (發散)}\end{aligned}$$

▶ $$\begin{aligned}\int_{-1}^0 \frac{1}{x^2}\,\mathrm{d}x &= \lim_{t\to 0^-}\int_{-1}^t x^{-2}\,\mathrm{d}x\\ &= \lim_{t\to 0^-}\left[-x^{-1}\right]_{-1}^t\\ &= \lim_{t\to 0^-}\left(-\frac{1}{t}-1\right) = \infty \text{ (發散)}\end{aligned}$$

Theorem. (**直接比較審斂法 (Direct Comparison Theorem)**) 給定連續函數 $f(x)$、$g(x)$，對於任意 $x\ge a$ 有 $f(x)\ge g(x)\ge 0$，那麼：

1. 若 $\displaystyle\int_a^\infty f(x)\,\mathrm{d}x$ 收斂，則 $\displaystyle\int_a^\infty g(x)\,\mathrm{d}x$ 將 收斂
2. 若 $\displaystyle\int_a^\infty g(x)\,\mathrm{d}x$ 發散，則 $\displaystyle\int_a^\infty f(x)\,\mathrm{d}x$ 將 發散

Question 判斷以下瑕積分是否收斂.

▷ $\displaystyle\int_1^\infty \frac{\sin^2 x}{x^2}\,\mathrm{d}x$ (Hint: $0\le \sin^2 x\le 1, \forall x\in\mathbb{R}$)

由於 $0\le \sin^2 x\le 1$, 我們有 $\displaystyle 0\le \frac{\sin^2 x}{x^2}\le \frac{1}{x^2},\quad \forall x\ge 1$

並且積分 $\displaystyle\int_1^\infty \frac{1}{x^2}\,\mathrm{d}x = 1$ 收斂

由直接比較審斂法，$\displaystyle\int_1^\infty \frac{\sin^2 x}{x^2}\,\mathrm{d}x$ 亦收斂.

附錄：積分表

指數與三角函數

$$\int e^{ax}\,\mathrm{d}x = \frac{1}{a}e^{ax} + C \qquad \int \sin(ax)\,\mathrm{d}x = -\frac{1}{a}\cos(ax) + C$$

$$\int \cos(ax)\,\mathrm{d}x = \frac{1}{a}\sin(ax) + C \qquad \int \tan(ax)\,\mathrm{d}x = \frac{1}{a}\ln|\sec(ax)| + C$$

$$\int \sec(ax)\,\mathrm{d}x = \frac{1}{a}\ln|\sec(ax)+\tan(ax)| + C \qquad \int \csc(ax)\,\mathrm{d}x = \frac{1}{a}\ln|\csc(ax)-\cot(ax)| + C$$

$$\int \sin^2(ax)\,\mathrm{d}x = \frac{x}{2} - \frac{\sin(2ax)}{4a} + C \qquad \int \cos^2(ax)\,\mathrm{d}x = \frac{x}{2} + \frac{\sin(2ax)}{4a} + C$$

有理函數與部分分式型

$$\int \frac{1}{x+a}\,\mathrm{d}x = \ln|x+a| + C \qquad \int \frac{f'(x)}{f(x)}\,\mathrm{d}x = \ln|f(x)| + C$$

$$\int \frac{1}{x^2+a^2}\,\mathrm{d}x = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C \qquad \int \frac{x}{x^2+a^2}\,\mathrm{d}x = \frac{1}{2}\ln(x^2+a^2) + C$$

$$\int \frac{1}{x^2-a^2}\,\mathrm{d}x = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C \qquad \int \frac{1}{(x+a)^n}\,\mathrm{d}x = \frac{-1}{(n-1)(x+a)^{n-1}} + C$$

衰減震盪型

$$\int e^{ax}\sin(bx)\,\mathrm{d}x = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx) + C$$

$$\int e^{ax}\cos(bx)\,\mathrm{d}x = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + C$$

Table 1: 常用積分表

歷屆試題

▶ 積分技巧與基本計算

1. (師大微乙 (一)114#3) Evaluate the integrals. (計算積分)

   (a) $\displaystyle\int_0^{\pi/2} \cos^3(3\theta)\sin^2(6\theta)\,d\theta$

   (b) $\displaystyle\int x^3\sin(2x)\,dx$

   (c) $\displaystyle\int \frac{2x^2-6x+4}{(x^2+1)(x-1)^3}\,dx$

   (d) $\displaystyle\int \tan^{-1}(\sqrt{x})\,dx$

   (e) $\displaystyle\int \sqrt[3]{\tan x}\,dx$

   (f) $\displaystyle\int \ln(x)\cdot x^{\ln(x)-1}\,dx$

   (g) $\displaystyle\int_0^{2025} \frac{\sqrt{x}}{\sqrt{x}+\sqrt{2025-x}}\,dx$

   (h) $\displaystyle\int \frac{\sin^2(x)}{1+\cos^2(x)}\,dx$

2. (師大微乙 (一)113 暑 #2) Evaluate the following integrals.

   (a) $\displaystyle\int \left(x+\sqrt{x}+1+\frac{1}{\sqrt{x}}+\frac{1}{x}\right)\,\mathrm{d}x$

   (b) $\displaystyle\int \left(x^2\sin x+\cos^2 x\right)\,\mathrm{d}x$

   (c) $\displaystyle\int \left(\tan^2 x+\tan^4 x\right)\,\mathrm{d}x$

   (d) $\displaystyle\int \frac{2x}{\sqrt{1+x^4}}\,\mathrm{d}x$

   (e) $\displaystyle\int \frac{e^x}{e^{2x}-3e^x+2}\,\mathrm{d}x$

   (f) $\displaystyle\int_{-1}^{1}(x^3+3)\sqrt{4-x^2}\,\mathrm{d}x$

3. (師大微乙 (一)113 暑 #3) Let a be a real number and $a>0$.

   (a) Evaluate $\displaystyle\int x\ln x\,\mathrm{d}x$

   (b) If a satisfies the equation $\displaystyle\int_0^a x\ln x\,\mathrm{d}x=0$, find the value of a.

4. (師大微乙 (一)113#2) Evaluate the integrals.

   (a) $\displaystyle\int \frac{\cos^5 x}{\sin^2 x}\,\mathrm{d}x$

   (b) $\displaystyle\int \frac{x^2}{\sqrt{49-x^2}}\,\mathrm{d}x$

   (c) $\displaystyle\int \frac{2x^3+4}{(x-1)^2(x^2+1)}\,\mathrm{d}x$

   (d) $\displaystyle\int \frac{\tan x(\sec^2 x+4\sec x)}{\sec^2 x+5\sec x-6}\,\mathrm{d}x$

   (e) $\displaystyle\int_{-1}^{1}\sqrt{x^{10}}+x^2\sec x\tan x+\frac{1}{1+x^2}\,\mathrm{d}x$

5. (師大微乙 (一)113#6)

   (a) Evaluate $\displaystyle\int_0^1 \ln x\,\mathrm{d}x$. (Note: this integral is an improper integral(瑕積分).)

   (b) Let a be a real number and $a>0$. If a satisfies the equation $\displaystyle\int_0^a \ln x\,\mathrm{d}x=0$, find a.

6. (師大微乙 (一)112#2) Evaluate the following integrals.

(a) $\displaystyle\int \frac{x^2}{x^4-1}\,\mathrm{d}x$

(b) $\displaystyle\int_0^{\pi^2/4} \cos\sqrt{x}\,\mathrm{d}x$

(c) $\displaystyle\int_0^{e} \ln x\,\mathrm{d}x$

(d) $\displaystyle\int \sec^5 x\,\mathrm{d}x$

(e) $\displaystyle\int \frac{(1-y^2)^{5/2}}{y^8}\,\mathrm{d}y$

7. (師大微乙 (一)111#1) Evaluate the integrals.

(a) $\displaystyle\int \frac{\ln(\ln x)}{x\ln x}\,\mathrm{d}x$

(b) $\displaystyle\int_0^1 x\ln x\,\mathrm{d}x$

(c) $\displaystyle\int \frac{x^3}{x^2+2x+1}\,\mathrm{d}x$

(d) $\displaystyle\int \frac{x}{\sqrt{1+x^4}}\,\mathrm{d}x$

(e) $\displaystyle\int \sqrt{\sin x}\cos^3 x\,\mathrm{d}x$

(f) $\displaystyle\int \frac{xe^{2x}}{(2x+1)^2}\,\mathrm{d}x$

(g) $\displaystyle\int \frac{\sec^2\theta}{\tan^3\theta-\tan^2\theta}d\theta$

(h) $\displaystyle\int_\infty^1 \frac{9r^2}{\sqrt{1-r^3}}\,\mathrm{d}r$

8. (師大微乙 (一)110#1) 計算下列積分. (Evaluate the following integrals.)

(a) $\displaystyle\int_1^4 (\frac{2}{x^3}-\sqrt{x})\,\mathrm{d}x$

(b) $\displaystyle\int x^2\sqrt{x^3+1}\,\mathrm{d}x$

(c) $\displaystyle\int_0^3 \frac{x}{\sqrt{x+1}}\,\mathrm{d}x$

(d) $\displaystyle\int \cos\sqrt{x}\,\mathrm{d}x$

(e) $\displaystyle\int e^x\cos 2x\,\mathrm{d}x$

(f) $\displaystyle\int \frac{2}{16x^2-1}\,\mathrm{d}x$

(g) $\displaystyle\int \sqrt{9-x^2}\,\mathrm{d}x$

(h) $\displaystyle\int_4^6 \frac{1}{\sqrt{8x-x^2}}\,\mathrm{d}x$

9. (師大微乙 (一)110#5) 計算以下瑕積分. (Evaluate the following improper integrals.)

(a) $\displaystyle\int_0^\infty xe^{-2x}\,\mathrm{d}x$

(b) $\displaystyle\int_0^1 \ln x\,\mathrm{d}x$

10. (師大微乙 (一)109#1) Evaluate the integrals.

(a) $\displaystyle\int \sqrt{16-4x^2}\,\mathrm{d}x$

(b) $\displaystyle\int_1^\infty \frac{1}{x(\ln x)^3}\,\mathrm{d}x$

(c) $\displaystyle\int x^4\cos(2x)\,\mathrm{d}x$

(d) $\displaystyle\int \sin(7x)\cos(3x)\,\mathrm{d}x$

(e) $\displaystyle\int \frac{\sqrt{x^2+3}}{x}\,\mathrm{d}x$

(f) $\displaystyle\int \sec(2x)\tan^2(2x)\,\mathrm{d}x$

(g) $\displaystyle\int_0^{\pi/2} \frac{\cos(\theta)}{(\sin^2(\theta)+4\cos^2(\theta))^{3/2}}d\theta$

(h) $\displaystyle\int \frac{5x^2+20x+6}{x^3+2x^2+x}\,\mathrm{d}x$

11. (師大微乙 (一)108#3)

    (a) Does the improper integral $\displaystyle\int_1^\infty \frac{\ln t}{2\sqrt{t}}\,\mathrm{d}t$ converge or diverge? Give reasons for your answer.

    (b) Use the part (a) to evaluate $\displaystyle\lim_{x\to\infty}\frac{\int_1^x \frac{\ln t}{2\sqrt{t}}\,\mathrm{d}t}{\sqrt{x^3}}$.

12. (師大微乙 (一)108#4) Find each integral.

    (a) $\displaystyle\int \frac{8x}{x^3+x^2-x-1}\,\mathrm{d}x$

    (b) $\displaystyle\int \tan^4(5x)\,\mathrm{d}x$

    (c) $\displaystyle\int e^x\sqrt{1-e^{2x}}\,\mathrm{d}x$

    (d) $\displaystyle\int e^x\cos x\,\mathrm{d}x$

    (e) $\displaystyle\int_0^{\pi/2}\frac{\cos^5 x}{\sin^5 x+\cos^5 x}\,\mathrm{d}x$ (Hint: Use the substitution $u=\frac{\pi}{2}-x$.)

13. (師大微乙 (一)107#1) Evaluate the integrals.

    (a) $\displaystyle\int \frac{2x+1}{\sqrt{4-x}}\,\mathrm{d}x$

    (b) $\displaystyle\int \frac{1}{\sqrt{x}(1+\sqrt{x})}\,\mathrm{d}x$

    (c) $\displaystyle\int xe^{2x}\,\mathrm{d}x$

    (d) $\displaystyle\int \frac{-x^3+5x^2-4x+9}{x^4-2x^3+3x^2-4x+2}\,\mathrm{d}x$

    (e) $\displaystyle\int \frac{x}{x^2-2x+3}\,\mathrm{d}x$

    (f) $\displaystyle\int \tan x\cdot\cos(2x)\,\mathrm{d}x$

    (g) $\displaystyle\int_0^{\sqrt{3}}\frac{x^2}{(4-x^2)^{3/2}}\,\mathrm{d}x$

    (h) $\displaystyle\int \sqrt{1+x^{-1}}\,\mathrm{d}x$

14. (師大微乙 (一)107#5) Find the improper integral

$$\int_0^4 \frac{\ln x}{\sqrt{x}}\,\mathrm{d}x.$$

15. (師大微乙 (一)106#1) Evaluate the integrals.

    (a) $\displaystyle\int (x+1)5^{(x+1)^2}\,\mathrm{d}x$

    (b) $\displaystyle\int \frac{2x^3-4x^2-15x+5}{x^2-2x-8}\,\mathrm{d}x$

    (c) $\displaystyle\int e^{\sqrt{2x}}\,\mathrm{d}x$

    (d) $\displaystyle\int e^x\sqrt{1-e^{2x}}\,\mathrm{d}x$

    (e) $\displaystyle\int \sec^4 x\tan^3 x\,\mathrm{d}x$

    (f) $\displaystyle\int \frac{\sin x}{\cos x-\cos^2 x}\,\mathrm{d}x$

    (g) $\displaystyle\int \frac{\ln\sqrt{x}}{x}\,\mathrm{d}x$

    (h) $\displaystyle\int_0^1 x\ln x\,\mathrm{d}x$

16. (師大微乙 (一)106#5) 給定下列之瑕積分 (improper integral)，請做以下兩小題.

$$\int_1^\infty (\frac{c}{x^2}-\frac{1}{3x})\,\mathrm{d}x$$

(a) 找出使得上述瑕積分收斂 (converge) 的 c 值.

(b) 求 (a) 答案 c 值對應的之瑕積分值.

17. (師大微乙 (一)105#1) Evaluate the integrals.

(a) $\displaystyle\int_1^4 (3-|x-3|)\,\mathrm{d}x$

(b) $\displaystyle\int \frac{3}{x^2+3x-10}\,\mathrm{d}x$

(c) $\displaystyle\int \sin^2\theta\tan\theta\sqrt{\cos\theta}d\theta$

(d) $\displaystyle\int \frac{\sec x\tan x}{2\sec x-1}\,\mathrm{d}x$

(e) $\displaystyle\int \frac{\ln x}{x^3}\,\mathrm{d}x$

(f) $\displaystyle\int \frac{\sqrt{x^2-25}}{x}\,\mathrm{d}x$

(g) $\displaystyle\int \frac{x^3e^{x^2}}{(x^2+1)^2}\,\mathrm{d}x$

(h) $\displaystyle\int_0^2 \frac{1}{\sqrt[3]{x-1}}\,\mathrm{d}x$

18. (師大微乙 (一)104#1) 計算下列積分：

(a) $\displaystyle\int_0^1 (x^3+2x)\,\mathrm{d}x$

(b) $\displaystyle\int \frac{2x+1}{x^2-7x+12}\,\mathrm{d}x$

(c) $\displaystyle\int_0^{\pi/2} \cos^3(x)\,\mathrm{d}x$

(d) $\displaystyle\int_1^e \frac{\mathrm{d}y}{y(1+(\ln y)^2)}$

(e) $\displaystyle\int \sec^3(z)\,\mathrm{d}z$

(f) $\displaystyle\int_1^9 \frac{\log_3 x}{x}\,\mathrm{d}x$

(g) $\displaystyle\int \sin\sqrt{2x}\,\mathrm{d}x$

(h) $\displaystyle\int \frac{\mathrm{d}s}{1+e^s}$

19. (師大微乙 (一)104#5) 判斷下列瑕積分收斂或發散 (請標明使用何種檢定法).

$$\int_1^\infty \frac{1}{\sqrt{x^6+1}}\,\mathrm{d}x$$

20. (師大微乙 (一)103#1) 底下共有 8 題積分計算題

(a) $\displaystyle\int_1^4 x^2+3x+1\,\mathrm{d}x$

(b) $\displaystyle\int \frac{1}{x^2-5x+6}\,\mathrm{d}x$

(c) $\displaystyle\int \sin^3 x\cos^4 x\,\mathrm{d}x$

(d) $\displaystyle\int \tan^2 x\sec^3 x\,\mathrm{d}x$

(e) $\displaystyle\int \frac{\sqrt{1-x^2}}{x}\,\mathrm{d}x$

(f) $\displaystyle\int \ln x\,\mathrm{d}x$

(g) $\displaystyle\int \frac{\ln x}{x}\,\mathrm{d}x$

(h) $\displaystyle\int \frac{2z}{\sqrt[3]{z^2+1}}\,\mathrm{d}z$

21. (師大微乙 (一)103#5)

(a) 驗算 $\displaystyle\int_0^2 x^2\,\mathrm{d}x+\int_0^4 \sqrt{x}\,\mathrm{d}x=8$

(b) 求 $\int_0^{\frac{\pi}{2}} \sin x \, \mathrm{d}x + \int_0^1 \sin^{-1} x \, \mathrm{d}x$

(d) 判斷下列瑕積分收斂或發散，若收斂請計算其值.

$$\int_0^\infty \frac{e^{\tan^{-1} x}}{1+x^2} \, \mathrm{d}x$$

22. (師大微乙 (一)102#1) 計算下列積分:

(a) $\int_0^1 (x^2+3x+2) \, \mathrm{d}x$

(b) $\int \frac{1}{x^2+3x+2} \, \mathrm{d}x$

(c) $\int_0^1 x\sqrt{1-x} \, \mathrm{d}x$

(d) $\int_0^{2/3} \sqrt{4-9x^2} \, \mathrm{d}x$

(e) $\int \tan^3 x \sec^4 x \, \mathrm{d}x$

(f) $\int \frac{1+\sin x}{\cos^2 x} \, \mathrm{d}x$

(g) $\int x^2 e^{-x} \, \mathrm{d}x$

(h) $\int \sin\sqrt{x} \, \mathrm{d}x$

23. (師大微乙 (一)102#5) 判斷下列瑕積分收斂或發散，若收斂，計算其值.

(a) $\int_0^\infty \frac{1}{x^2+1} dx$

(b) $\int_0^\infty \frac{1}{\sqrt{x}+1} dx.$